

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy .

Reasoning + Answer:

1. We are given the following logarithmic expressions involving real numbers x and y , both greater than 1:

$$\log_x(y^x)=10 \quad \text{and} \quad \log_y(x^{4y})=10$$

We need to find the value of xy .

2. **First expression:**

$$\log_x(y^x)=10$$

Use logarithmic identities:

$$\log_x(y^x)=x \log_x(y)$$

Given that this equals 10:

$$x \log_x(y)=10$$

Now, recall that $\log_x(y)=\frac{\log y}{\log x}$. So,

$$x \cdot \frac{\log y}{\log x}=10$$

This can be rewritten:

$$\frac{x \log y}{\log x}=10$$

3. **Second expression:**

$$\log_y(x^{4y})=10$$

$$4y \log_y(x)=10$$

Again, $\log_y(x)=\frac{\log x}{\log y}$, so:

$$4y \cdot \frac{\log x}{\log y}=10$$

Therefore,

$$\frac{4y \log x}{\log y}=10$$

4. **We now have equations (1) and (2):**

$$\frac{x \log y}{\log x}=10 \quad \text{(from equation 1)}$$

$$\frac{4y \log x}{\log y}=10 \quad \text{(from equation 2)}$$

5. Let's simplify both equations.

From equation (1):

$$\frac{x}{\log x / \log y}=10 \Rightarrow \frac{x}{\log x / \log y}=10 \Rightarrow \frac{x \log y}{\log x}=10$$

So we can write this as:

$$\frac{x}{\log x} \cdot \log y=10$$

Define $a=\log x$ and $b=\log y$. Now, our equations become:

- From equation (1): $\frac{x^b}{a}=10$
- From equation (2): $\frac{4y^a}{b}=10$

But $x=10^a$ and $y=10^b$, because $a=\log x \Rightarrow x=10^a$, etc.

So substitute $x=10^a$ and $y=10^b$ into both:

- First equation:

$$\frac{10^{a \cdot b}}{a}=10 \Rightarrow \frac{b \cdot 10^a}{a}=10$$

- Second equation:

$$\frac{4 \cdot 10^b \cdot a}{b}=10 \Rightarrow \frac{4a \cdot 10^b}{b}=10$$

$$\tag{4}$$